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Vision Based Automation system for Onion Grading Using Robotic Arm

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***Abstract*—In agriculture , onion grading is a essential process to ensure the quality is maintained. The methods implemented nowadays are rudimentary and consist of manual checks that are labour intensive and prone to inconsistencies and inaccuracies. The system consists of a conveyor belt , proximity sensors , a high resolution camera mounted on a robotic arm , Programmable Logic Controller and an intelligent grading algorithm developed using YOLOv8 & CNN to classify onions into 2 types: Good and Bad Onions , based on parameters such as shape , size , surface defects. There are 2 proximity sensors placed on extreme ends of the conveyor . The First sensor detects the onion and starts the conveyor while the second sensor which is placed at the end of the conveyor detects the onion and stops the conveyor and gives an activation signal to the robotic arm . The robot is equipped with a Luxonis Oak 1 Lite monocular DepthAI camera along with a suction gripper. The robotic arm then moves to the inspection point and runs our onion quality detection algorithm using the Oak 1 Lite camera. Based on these results , the robot segregates the onions into their respective boxes. The entire system is controlled by the PLC and the robot is programmed by a personal computer using UART Communication protocol. The entire system is designed to minimise human intervention in the entire process. This paper presents the system development methodology, including hardware integration, algorithm development and implementation. The experiments conducted on the system demonstrate the ability to detect and segregate onions effectively. The following proposed automation system is an easy integrable solution for the agricultural industry and paves the way for future innovations in the agricultural sector.**

***Index Terms*—Onion grading, automation system, agricultural automation, image processing, automation in agriculture.**

I. Introduction

In the agricultural sector, there are lot of inconsistencies and inaccuracies during manual grading of onions and it is important to ensure that the grading is consistent and efficient for quality control and optimize the sorting process therefore meeting the market standards. Onion grading is a key step after the post harvest process, traditionally relies on manual methods, which are time-consuming and labor-intensive. With the increase in demand there is a need for precision and efficiency in food production, automation plays a pivotal role and has become a key solution in improving the operational workflow in agriculture [3], [6], [12], [13].

The focus of this research is to develop an automation system for grading of onions such that it will be more accurate and precise than the existing system as well as scalable. The proposed system combines an intelligent grading algorithm, a high resolution camera which is mounted on a robotic arm, two proximity sensors, and a conveyor system to classify onions into Good Onions and Bad Onions. They are categorized into good and bad through assessments based on parameters such as size, shape, and surface imperfections.

This Research paper presents a comprehensive overview of the development and implementation of the automated system for grading of onions. The system uses a conveyor to transport onions to detection point, where they are captured by the high quality camera which is mounted on the robotic arm, where they are captured by the high-quality camera mounted on the robotic arm. The captured images are then processed by a grading algorithm that assign each onion to one of two catgories- Good or Bad. Following the detection the robotic arm picks the onion using vaccum gripper and drops the onion in the designated bin. The robot operates via UART communication, while the overall system is managed using a Programmable Logic Controller (PLC).

By introducing an automation solution solution to onion grading, this research seeks to address the challenges of manual grading while paving the way for more efficient, scalable, and precise agricultural practices. The implementation of such a system not only enhances productivity but also contributes to the growing field of precision agriculture, where technology-driven solutions are transforming traditional farming methods.

This paper is structured as follows:

- *Section One:* Introduction
- *Section Two:* Literature Review
- *Section Three:* Methodology
- *Section Four:* Result and Discussion
- *Section Five:* Conclusion

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II. Literature Review

The grading of agricultural produce, especially onions, has improved through different technological approaches, thereby increasing general productivity and efficiency. This study addresses the most important approaches in automated grading systems, including machine learning, deep learning, computer vision, sensor integration, and automation.

1) Artificial Intelligence in Agricultural Grading: The current state of agricultural product grading is profoundly advanced with the implementation of artificial intelligence and deep learning models. A variety of crops such as dragon fruit have been graded using techniques that include Support Vector Machines (SVM), Random Forests, and AI. These models indicate that similar techniques could be employed for onion grading where precise image data classification and sorting is possible. Advanced deep learning models such as ResNet, VGG, GoogLeNet, and EfficientNet have tremendous potential and ability to determine fruits classification and freshness grade, and hence can be used for onion grading. These models are directly applicable to onion grading since they incorporate real-time and high-precision technology. In addition, they enhance general grading systems by picture augmentation methods that raise the accuracy of classification [1], [10], [11], [16], [17].

2) Computer Vision and Object Image Processing: When identifying features like size, color, and shape in products, advanced methods like computer vision and image processing are crucial for sorting. Research on fruit sorting provides helpful ideas that can be applied to sorting onions. For example, studies have examined and showed how to determine a fruit by its size and color, which can also help to improve the onion grading. Some of the techniques like x-ray and near-infrared (NIR) spectroscopy which are non destructive techniques are been used to detect the hidden inner defects that are not possible to been by human eye. Since the inner qulaity of onions affect their market value significantly, these techniques are very important for evaluation. Additionally, using sound based methods such as ultrasound to assess the quality of fruits and vegetables is becoming more popular. This expands the horizon for reliable options which will help in onion sorting [5], [6], [15], [22], [24], [25], [26].

3) Sensor Integration in Automated Grading: Grading Systems consist of key components such as Gas Sensors , Temperature sensors , ultrasonic sensors used in agriculture and industry. These sensors gather data in real time , crucial for segregating produce based on their characteristics such as shape, size and quality. Utilization of ultrasonic sensors speeds up the grading process to a capacity of three tons per hour. Research shows that defects in onions such as internal rotting can be identified using special light transmitting systems. This showcases significant improvements and developments, sensors contribute to automated grading [2], [3], [7], [8], [23].

4) Advancements in Grading Systems: Manually sorting processes executed by humans are often slow and prone to error resulting in higher cost and less efficiency. Automated Grading Systems enhance the speed, accuracy and simultaneously brings down labor costs in grading process. Motorized systems are capable of handling substantial amount of produce with minimal human prevention , proving ideal for onion sorting systems. Benefits of these automated systems include Handling and sorting of up to 1000 onions an hour. This highlights how automation in agriculture leads to increased cost savings and production as machines operate more efficiently and quickly compared to human operators [12], [20], [21].

5) Bibliometric Evaluation and Future Research: Bibliometric analysis and evaluation in research give comprehensive insights that exist in current automated grading systems. By examining the past studies , researchers learn and understand all bout the successful methods implemented, in combination with the latest technologies available. Sensor Technology , Deep Learning and Machine Learning are often used to tackle challenges encountered during grading. Insights into these key technologies ensures continuous innovation and integration of systems that can keep up with the growing demand of the agriculture demand of the rising demand of the agriculture market [4], [9], [18], [19].

6) Emerging Trends: There have been continuous advancements in automation and AI, onion grading still runs into various challenges. Similar to other horticultural produce, onions are very perishable, and handling, storage, and shipping conditions may impact their quality. These concerns need to be addressed by the existing grading schemes without compromising on high speed and accuracy. Literature reviews indicate that better systems are needed to accommodate large-scale farming activities in developed and developing countries. To accommodate high-output farming without compromising uniform quality control processes, future studies need to focus on making the grading systems more scalable and flexible [13], [14].

III. Methodology

A. System Overview

The system comprises the following units:

1) **Input Stage**
2) **Detection Stage**
3) **Sorting Stage**

Initially, onions are placed on a feeder from where they fall on the conveyor system. The Conveyor System Consists of two proximity sensors; the first proximity sensor detects the presence of onion at the start of the conveyor and as soon as the onions are detected the sensor sends the signal to the Programmable Logic Controller (PLC), which then starts the conveyor; similarly, the second depth sensor detects the presence of onions at the end of the conveyor and when detected sends the signal to PLC which then stops the conveyor.As soon as the conveyor stops, the robot gets the signal from PLC using Modbus Communication. Then the robot moves to a detection point which has been pre-programmed and starts the detection. The DepthAI Camera captures the images and then the YOLOv8 Model detects the onions in that image, then

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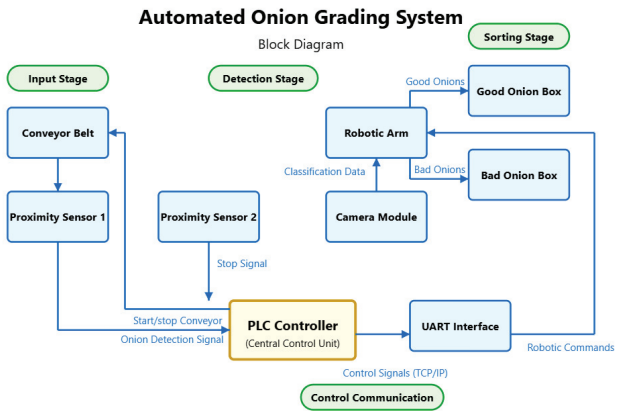


Fig. 1. Automated Onion Grading System

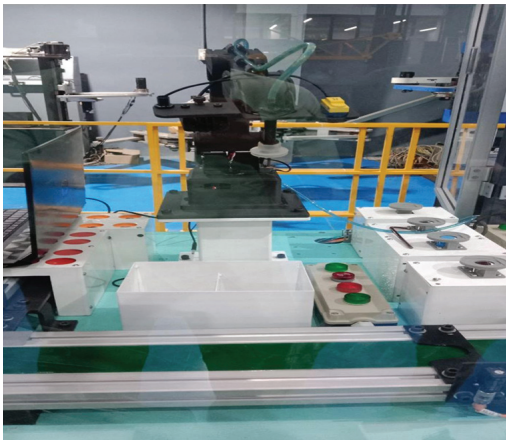


Fig. 2. Onion Grading System

the CNN model extracts the features from the image such as size, shape of onions, and classifies them accordingly. After classifying, the robot picks up the onions using the Vaccum gripper and places them in its respective box.

B. Robot Manipulation and Sorting Strategy

The system combines AI detection capabilities with robotic manipulation to perform precise and efficient sorting operations. The robotic arm which includes 4 degrees of freedom (DOF) functions as the primary component to carry out pick-and-place operations using outputs from AI model classification. A connection which enables continuous communication between detection system components emerges through the combination of DepthAI camera and ESP32. The DepthAI camera gathers image from the conveyor belt before processing them with YOLOv8 model for detection alongside the CNN model execution of defect classifications. After the AI models identify the onion type then it sends the results to the robotic control system that runs the obtained results through interpretation to execute appropriate motion commands.

The robotic arm executes correct movements through an inverse kinematics-based motion planning sequence. The required joint angles and end-effector position are computed by the system to help the robotic arm reach the target location without extra positional modifications after an onion classification. The robotic arm then places the onion directly in the designated Box.

C. Novelty of the concept

The proposed system employs advanced and novel approach by integrating image processing techniques as well as machine learning methods to develop an improved grading system for onions with enhanced accuracy along with efficiency benefits. The system uses a DepthAI camera to gather real-time onion images from conveyor belt. In contrast with traditional methods it avoids relying on manual inspection or basic sensor-based systems. The images undergo preprocessing for optimization. The object detection capabilities of YOLOv8 perform exact location and identification of onions during the process. Additionally, the system implements Convolutional Neural Networks (CNNs) to extract size-related and shape characteristics along with color-related defect features, ensuring accurate classification and grading. This integration creates reliable operation with scalability features that establish it as an innovative solution for automated onion sorting [27].

IV. Result and Discussion

1) System Performance Evaluation: The evaluation of the automated onion grading system took place through five critical stages consisting of image capture and preprocessing along with object detection and feature extraction before grading and sorting occurred. Tests were performed systematically for each processing procedure to confirm both reliable performance and efficient operation in the classification system for onions.

The DepthAI camera obtained high-quality images from the camera feed which provided suitable inputs for processing needs. The detection module based on YOLOv8 performed object detection with high precision to identify onions in the analyzed images. The system accurately identified the onions in the processed images which proves the detection model correctly distinguishes between onions and other background pixels.

The CNN-based feature extraction process aimed to find essential components necessary for onion quality assessment during the feature extraction stage. The model achieved precision in its ability to detect visual features which included form and surface quality and appearance detail. Training the system with more data would deliver improved performance in detecting subtle imperfections on the surface of the onions.

At the end extraction process is done to find important components necessary to classify onions. This results in accurate classification. The defined set of criteria such as size and shape allowed grading procedures to classify resulting in objective standardized assesments. All components of the automation system for grading of onions passed their evaluation tests resulting in adequate performance

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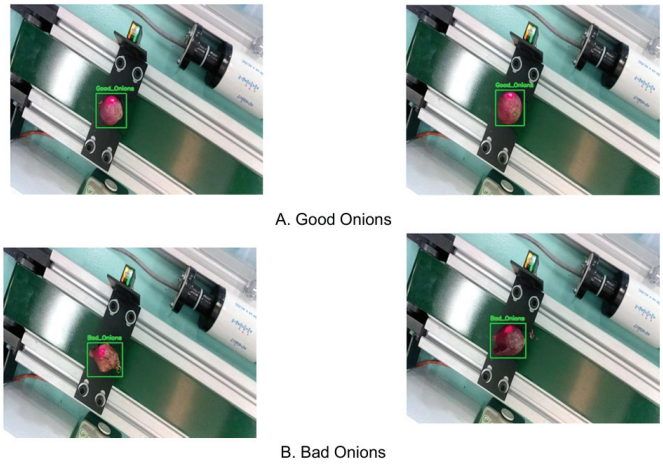


Fig. 3. Detection Results

The integrated system showcased enough functionality as each part efficiently passed its evaluation tests. The future work should be directed towards two main improvements that are better feature extraction and optimized detection to achieve enhanced grading accuracy.

TABLE I
SYSTEM PERFORMANCE PARAMETERS

Sr. No	Description	Input	Output	Pass/Fail	Remarks
1	Image Capture	Image feed	Captured image	Pass	Image captured successfully
2	Image Preprocessing	Raw image	Processed image	Pass	Image preprocessing successful
3	Object Detection (YOLOv8)	Processed image	Detected onions	Pass	Onions detected correctly
4	Feature Extraction (CNN)	Detected image	Extracted features	Pass	key features Detected
5	Grading	Processed image	Graded onions	Pass	Grading accurate

2) Model Performance Analysis: Figure 4 shows the average precision (mAP50) for the classification of good and bad onions using the trained model. The overall mAP score indicates the model's effectiveness in identifying the difference between different quality levels of onions.

Figure 5 illustrates the performace of YOLOv8 model in identifying and classifying between good and bad onions based on multiple parameters and evaluation metrics, including precision, recall, mAP50, and mAP50-95. The model performed best for good onions meanwhile its performance declined slightly for bad onions, it also also shows the traning and validation metrics of the CNN model over multiple epochs. For



Fig. 4. Model Performance on Validation Set: Average Precision by Class (mAP50)

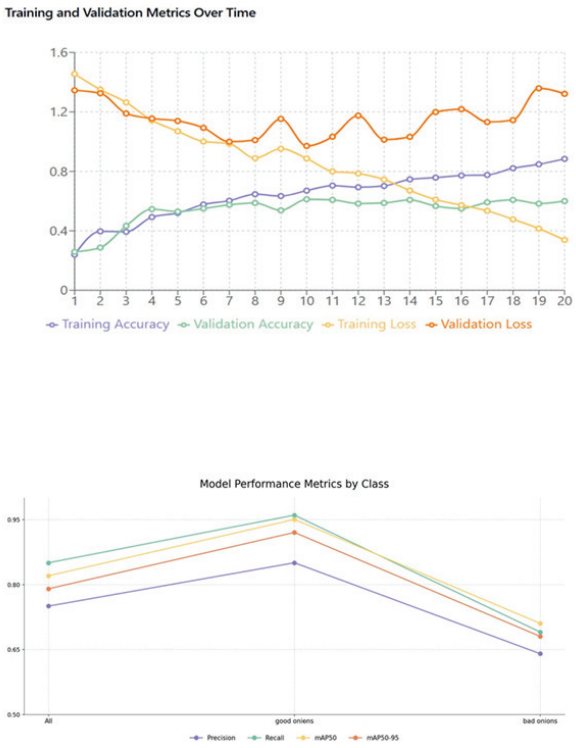


Fig. 5. CNN Training & YOLOv8 Performance Metrics for Onion Classification

CNN Training and validation accuracy increased over time, while training loss decreased, indicating learning progress. However, validation loss fluctuates, suggesting there is potential overfitting in later epochs.

The evaluation of YOLOv8 and CNN models was conducted to measure their effectiveness in accurately classifying onions within the system. The performance metrics analyzed included accuracy, precision, recall, and mean Average Precision (mAP). The YOLOv8-based detection model demonstrated strong detection capabilities, achieving a mAP of 85.5%, indicating a high level of confidence in correctly identifying and classifying onions. Subsequently, the precision was noted to be at 78.4%, reflecting the model's ability to reduce false

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positives, whereas the recall stood at 80.0%, showcasing its effectiveness and capability in correctly detecting all relevant objects.

V. Conclusion

The research aimed at developing an automation system for onion grading which uses advance AI models and a robotic solution to refine the sorting process. In this system we integrated a DepthAI camera, two proximity sensors, a conveyor system, and a 4-DOF robotic arm, the system was able to effectively classify onions.

This was achieved successfully by training and implementing two AI models. The first model was based on YOLOv8 and was responsible for detecting onions while the second model was developed using a Convolutional Neural Network (CNN) and was used for extracting and analyzing features to determine whether an onion is good or bad. Based on its assessment, the robotic arm places onions in its designated Box.

The model's performance was thoroughly analyzed by its accuracy and loss metrics over multiple epochs were considered while evaluating. The CNN based model showed effective training patterns in classifying and YOLOv8 model displayed progressive improvements in the detection accuracy.

The System was tested and its performance was evaluated based on classification accuracy, and robotic arm's repeatability and precision in sorting them. The results indicated that the proposed system notably reduces the rate of misclassification and increases the grading efficiency compared to conventional manual methods. By automating the grading process for onions, the system improves overall accuracy in grading while maintaining consistency and ensuring the onions meets the quality standards.

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